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Sigma Home: An IoT-based Home Automation using Node MCU

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***Abstract* - IoT (Internet of Things) based home automation systems have emerged and developed in recent years to remotely monitor and manage various appliances and gadgets. IoT can be connected through the internet as it is a device that helps make connections and monitor various devices and sensors. The paradigm proposed by us uses Node MCU which is an open source Wi-Fi-enabled microcontroller based on the ESP8266 chip, with IoT because it is simple and easy to interface with the network for home automation. To monitor various home devices like lights, temperature, humidity, and fan, different sensors are used. The data obtained from the sensors are then monitored through an Android application or a web-based interface, and with this the energy efficiency can be increased by controlling electrical fixtures through IoT. Here, the data recorded by sensors are fetched by the cloud for further analysis, perform different commands, and obtain intended result according to the requirement. This study proposes a novel system functioning, performance, and reliable solution for home automation. This system will make it a flexible and resourceful solution for IoT-based home motorization by providing a perfect experience with a minimum requirement of maintenance and also improving the quality of life by providing minimal cost, energy consumption, and sharp security. This research study focuses on the system in which multiple sensors are used and a layman can also get used to with this type of upcoming technology. Overall, the smart home automation system is scalable and provides a complete security to home, reduces human effort, and is also cost-effective.**

Keywords- IoT (Internet of Things), Home Automation, Node MCU, Wi-Fi, Android, Cloud, Microcontroller chip, ESP8266, Sensors.

1. INTRODUCTION

Nowadays, technology is increasing immensely to ease human efforts and make our life swifter and easier. The regular up-gradation and development in technology help us to dream of a world with all ease and comfort and all the work done automatically. Home automation is a perfect example depicting the technology which helps us in running all the household machines either simple or complex automatically [1].

IoT, the Internet of Things, is a platform that helps us interface physical machines with each other so that they can communicate among themselves, and this help in the automatic working of physical items without any direct human involvement [2]. This does not require any human-to-human or human-to-machine communication for the transfer of data over a network [3]. The network consists of a microprocessor and several sensors (Ultrasonic sensor, DHT22, Servo motor, LDR, PIR, etc.) among which the communication takes place. The purposed system helps the user to control all of their home appliances using the mobile application [4].

The IOT technology is increasing immensely day by day, this helps humans with easy automation in their homes. This technology is employed in our home for home automation as this helps in energy as well as time conservation which are major concerns of our generation as we are lacking both in today's busy schedule. As we can control all the home appliances within our palm and if there is no one in the home the PIR will sense that and brings the whole system to a shutdown mode, this helps in saving time and energy [4], [5].

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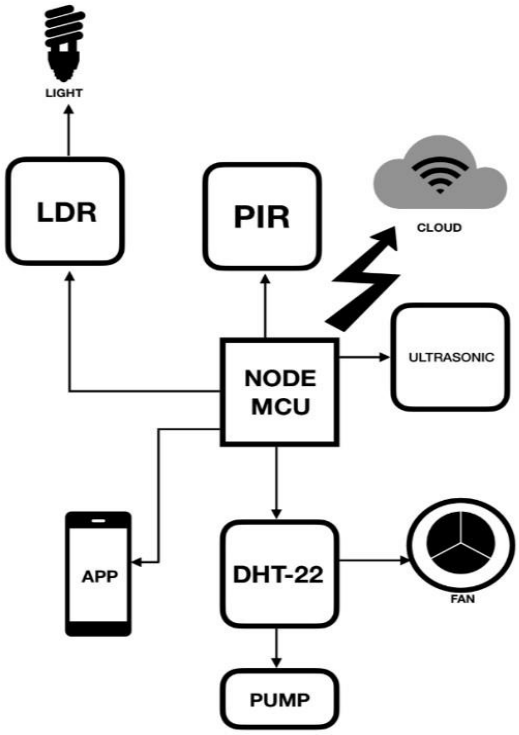

Fig. 1 Overview of the system

The main objective of the project is to construct a system which can easily make a home automatic and can control the basic appliances on its own, and to conserve energy.

The contribution of this is to provide an easy and comfortable life to all at a very less cost.

2. LITERATURE REVIEW

IoT-based home automation systems are getting increasingly famous due to the benefits they offer, which include increased convenience, improved electricity efficiency, and enhanced security. For home automation systems, node MCU is used to control and monitor various sensors and home appliances [6]. Node MCU is a user-friendly microcontroller that can interconnect with many sensors at the same time, making it more secure, cost-effective, and affordable [7]. It works effectively and efficiently for various tasks like lightning control, temperature monitoring, and security purposes by using the Android application or a smartwatch. In the system, many sensors have been used for performing different tasks such as PIR, Relay modules, DHT11, Servo motor, Ultrasonic, and LDR [8]. The sensors collect data and send that to the cloud for further processes. Blynk application is a mobile application that helps to add different controllers which can be used to control different sensors to do so we just need to connect the Wi-Fi module of Node MCU(ESP8266) with the mobile [1], [9]. The improvement in technology has completely changed the lives of humans by making it easier and effortless [10]. A smart security system includes sensors, and smart locks that can be controlled remotely and provide alerts if there is any unusual activity in the home, with this smart applications and voice recognition as well for controlling automation [11]. The trend of making a home automatic is increasing rapidly and due to this, every time new technology is discovered. Voice recognition technology is introduced to control the home just by giving voice commands. The voice assistant can understand commands because of their integration with home appliances, machine learning algorithms, and advancement in natural language processing. The voice assistant provides many benefits to the user by just controlling the devices and giving simple commands [12], [13]. A person can ensure total security of their home because of the home automation system and can control their home from anywhere from controlling the switches to watering the plants whenever required. The system has been tested and evaluated successfully and it has shown a great responsibility toward ensuring safety and improving energy efficiency [14]–[16]. Sensors work automatically and remotely like adjusting the temperature for a normal body and switching ON/OFF lights and fan accordingly to the availability of the person. As we are interfacing Node MCU with different sensors using the internet and if there is unavailability of the internet GSM (Global System for Mobile Communication) 3 technology can be used to work on the web services and we do not need to directly connect with the internet [17]. Generally, from existing literature and research we have come across that from home automation systems, we can remotely control the home appliances, run the devices at specific intervals of time, and can monitor the energy consumption time-to-time and from anywhere [18]. Through the study till now, we found that IoTbased Home Automation using Node MCU is in public demand to automate home appliances. A person can control his home from anywhere with just a click through his smartphone using a controlling application [19].

3. PROPOSED SYSTEM

The first requirement is to understand the need for home automation in everyday life. We need home automation to make people's lives comfortable, minimize the usage of energy, and ensure safety [15]. The proposed system consists of Node MCU and different sensors in the hardware part and an Android application or a web server and a user-friendly web interface, installed libraries for connecting sensors to Node MCU in the software part. We will connect the Wi-Fi module of Node MCU with the mobile and sensors, then send the data to the cloud. We will fetch the recorded data for further analysis and can monitor the home automatically. By using the Blynk application we can control different appliances. Users can control the home from their devices and ensure total security [3].

Fig. 2 explains the working of the project in which we first connect the Node MCU with Wi-Fi and then initialise the sensors. After that we read and analyse the sensors data and

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do any actions as per requirement. The sensors data is analysed through the cloud and referred through a set limit which is user defined.

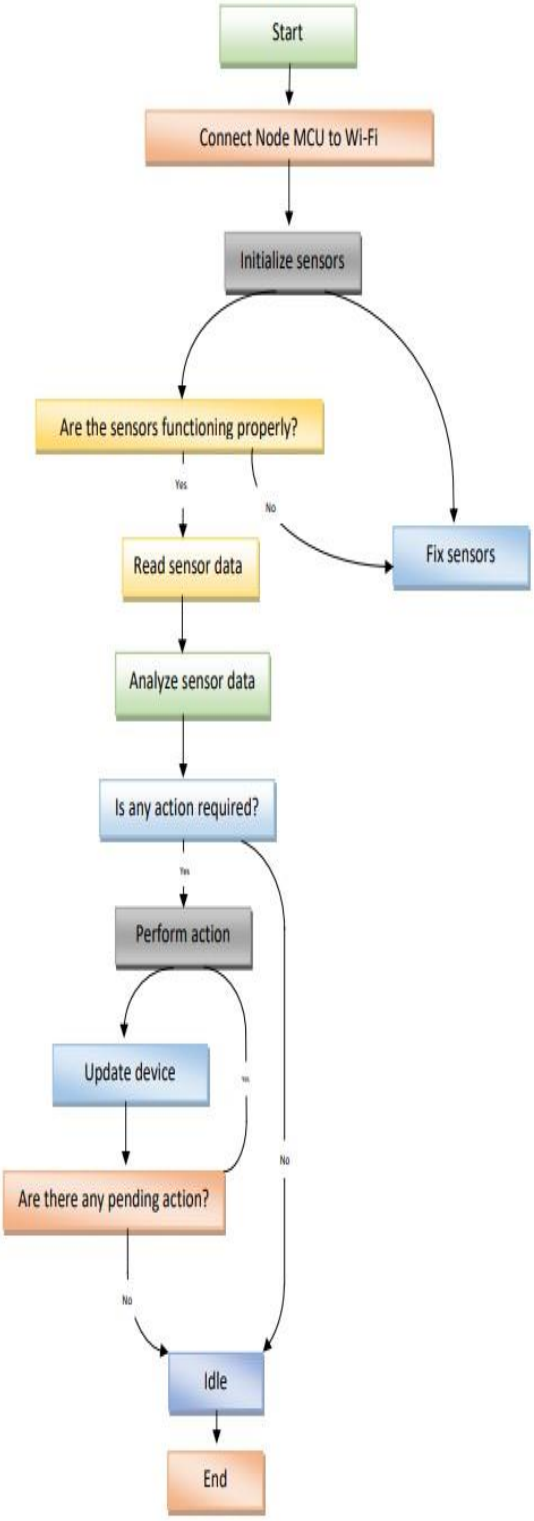

Fig. 2 Working of the System

4. IMPLEMENTATION

In our proposed work, we interfaced different hardware implementations with different software implementations. In the hardware part, we used Node MCU (ESP8266) microcontroller, and multiple sensors like PIR, LDR, DHT22, and Ultrasonic sensors. For the software, we first connect the Wi-Fi module of Node MCU with the personal network, integrate IoT with the cloud to share the recorded data, and control it accordingly. We also use an Android application, the Blynk app for controlling different appliances. The whole system provides an advanced home that is very convenient, energy-efficient, and secure.

A detailed description of different parts used in the system

A. Node MCU (ESP8266):

Node MCU is a development kit that provides an open-source platform that helps build sensor connections. The Wi-Fi module of Node MCU is connected to a hotspot of the device to perform further functioning. Node MCU which is an opensource Wi-Fi-enabled microcontroller based on the ESP8266 chip, with IoT because it is simple and easy to interface with the network for home automation [20]. It is used to control and monitor various sensors and home appliances. In our model, we used Node MCU to connect various sensors like DHT 22 sensor, ultrasonic sensor, LDR sensor, PIR sensor, and Relay. We are also connecting it with the Blynk application for controlling the sensors with the help of an inbuilt Wi-Fi module (ESP8266) present in Node MCU [14].

Node MCU is selected over other boards because it has inbuild wi-fi, it of low cost and it is very easy to use and it is of very small size. Overall, it satisfies all the requirement for the project.

Fig. 3 Node MCU

B. PIR sensor:

PIR (Passive Infrared sensor) is a 3-pin sensor that is used to detect motion. In this the VCC pin is connected to the 3V pin of the Node MCU, the ground pin is also connected to the ground pin of the Node MCU, and the output pin is connected to any one of the digital pins of the Node MCU. It is a passive device that detects infrared emitting radiation by humans and animals in motion [21]. It is used in automatic door openings and security systems. This sensor has a pyroelectric sensor

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and lens which works together for generating voltage and focusing on infrared radiation. It consumes less energy and is reliable and effective [7].

We are using it for the same application as mentioned above and also it is interfaced with LEDs in the staircase so that whenever there is a motion PIR will detect it and will switch on the light.

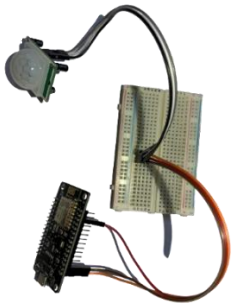

Fig. 4 Connection for PIR Sensor

C. Ultrasonic sensor:

HC-SR04 is the ultrasonic sensor having 4 pins named VCC, GND, Trig, and Echo. The VCC pin is connected to the 3V pin of Node MCU; the GND pin is connected to the ground pin of Node MCU; the Trig and Echo pin of the sensor is connected to the digital pins (let it be D3 and D4 respectively) of Node MCU. This sensor is used for detecting objects in front and also measuring the distance[22].

In our project, we have interfaced the ultrasonic sensor with the servo motor, and we have fixed a limit to this that whenever any object crosses that limit and comes closer to the sensor, the servo motor will get started. This is used to build the automatic door-opening system.

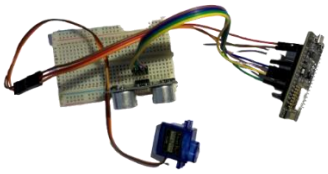

Fig. 5 Connection for Ultrasonic Sensor with Servo Motor

D. LDR:

LDR (Light Dependent Resistor) is a two-pin sensor whose negative pin is connected to the ground pin of Node MCU and the positive pin is connected to any of the digital pin of Node MCU. It is interfaced with LEDs for an automatic lighting system. Whenever there is darkness in the room the LDR sensor will detect it and will automatically switch on the LEDs and switch off them whenever there is sufficient light in the room[23]. We are also using an LDR sensor for the same purpose of automatic lighting as it will help in energy conservation because in day time the LDR sensor will automatically switch off all the light.

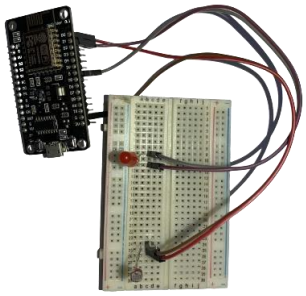

Fig. 6 Connection for LDR sensor

E. DHT-22:

DHT-22 is a 3-pin temperature humidity sensor that is used to detect temperature and humidity. In this, the positive and negative pin of the sensor is connected to the voltage pin and ground pin of Node MCU respectively and the output pin of the sensor is connected to any of the digital pin of Node MCU[7].

We interfaced it with the fan and exhaust fan so that whenever there is an increase in temperature and humidity in the room the sensor could detect it and when it passes the given limit it could which on the fan and exhaust respectively.

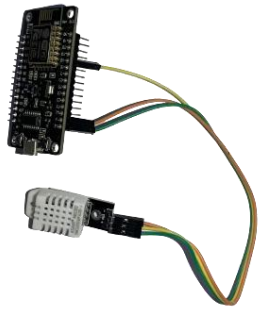

Fig. 7 Connection for DHT-22 sensor

F. Security Purpose:

For the security purpose, we are using two things

1. RF-ID.
2. Biometric (fingerprint sensor).

There are two keys for RF-ID and a desirable number of biometric allowances can be added according to user requirements. This method will provide double locking

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and double security to our main gate which is a main concern for any automated home.

G. *Blynk Application:*

The Blynk app is used for creating applications. It is an IOT interfacing medium used to interface the sensors and devices we use with our smartphones with the help of the internet. It has easy access and a variation of dashboards and we can create many web applications quite easily[24].

We are using Blynk 2.0 for our project through which we are interfacing our various sensors with our smartphones via the internet and we can easily control those sensors and devices using this app.

H. *IoT- Cloud Integration:*

IoT is used for linking and interchanging recorded data from different sensors on the cloud for further processing and controlling. IoT allows the cloud to increase the scope of dealing with real-life problems because of the advancement of technology. This system can be achieved at a low cost as it is effective and can be managed from anywhere [25].

5. RESULT

We have successfully designed a home automation system that is able to provide a better and comfortable life to the people living in it. We have interfaced different sensors together so that all the applications in the hoke can either be fully automatic or could be controlled from just a single mobile application.

Fig. 8 Connection of all sensors together

Fig. 9 Implementation of System with multiple sensors

6. CONCLUSION

In conclusion, an IoT-Based Home Automation using Node MCU is proven to be a better idea for making a home fully secure, a cost-effective solution for operating a home automatically. We first interfaced each module individually like the dooring system, lighting system, fan and exhaust system, and security. Then we interfaced all the modules and their programs together to make it a whole system or network.

We then connected our Node MCU with the mobile using the WIFI module and used the Blynk application for controlling our system. Home Automation is a very simple and easy process of just interfacing different sensors with an Android application using Node MCU and just by doing this, we can control fans, switches, air conditioners, lights, fans, and washing machines from anywhere. We can make an advanced home using Home Automation System which is better in every aspect than a normal home. This system will make it a flexible and resourceful solution for IoT-based home motorization by providing a perfect experience with a minimum requirement of maintenance and also improving the quality of life by providing minimal cost, energy consumption, and sharp security.

The main limitations which are highlighted are that the number of sensors attached is limited and the main focus of us is to increase that only, we are able to incorporate that somehow but that can be improved more by using customized boards rather than using market ready boards.

The performance can be improved by using the same customized boards to add more sensors.

7. FUTURE SCOPE

A smart home can be built with an in-built AI, which will leverage a better user experience. A camera can be installed to avoid intruders in the home when the owners are not available. The images from the camera can be uploaded to

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the cloud, which can be accessed by the owner whenever they feel like an intruder is trying to breach into their home. They can simply notify the system about it and the system will further notify the nearby police station about it with the images.

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